Owen's Math Test Review

Name: _____ Date: _____

Simple Solutions Standards-Based Mathematics 7 · Lessons #31, #32, #33 (book pages 62–67)

How to use this packet

1. Read the mini-lesson for each skill (pages 2–3). Try the example on scrap paper first.
2. Read "Fix-it" (page 4): four Lesson #31 problems to look at again, with the right way to do them.
3. Work all 36 test-review problems (Lessons #31–#33). Show your work in the space given.
4. Do the extra practice pages (A–E), which focus on the gold-star skills.
5. Check everything with the answer key at the back. Redo anything you missed, then try one more like it.
6. Bonus: the interactive page reads problems aloud, checks answers, and gives new problems for anything you miss.

Skills on the test

1. Scaling a recipe (proportions) ★	2. Inequalities ★
3. Decimals & money ★	4. Range, mode & mean ★
5. Unit rate & constant of proportionality	6. Fractions & complex fractions
7. Integers (add & multiply)	8. Expressions: simplify, evaluate, write
9. Equations	10. Volume & surface area

★ = focus skill (from Lesson #31). These get extra practice.

Which problems use which skill

- Scaling a recipe (proportions): L31-6, L32-5, L33-10
- Inequalities: L31-9, L32-7
- Decimals & money: L31-10, L31-12, L32-1, L32-11
- Range, mode & mean: L31-4
- Unit rate & constant of proportionality: L31-3, L31-7, L32-8, L33-8, L33-9
- Fractions & complex fractions: L31-1, L32-10, L33-1, L33-2
- Integers (add & multiply): L31-2, L31-5, L32-3, L32-12, L33-3, L33-6
- Expressions: simplify, evaluate, write: L31-8, L32-4, L32-6, L32-9, L33-4, L33-12
- Equations: L33-7, L33-11
- Volume & surface area: L31-11, L32-2, L33-5

Mini-Lessons & Tips

Each box: how to do it, an example, and something to remember.

1. Scaling a recipe (proportions) ★ FOCUS

L31-6, L32-5, L33-10

- Find the SCALE FACTOR first: new amount ÷ old amount (4 people ÷ 3 people = 4/3).
- Multiply EVERY ingredient by that same factor. Never add the same amount to each ingredient.
- Check: the ratio between ingredients should stay the same (2 cups nuts : 1 cup berries, so nuts = 2 × berries).

Example: Pancakes: 3 cups flour, 2 eggs, serves 4. For 6 people the factor is 6/4 = 3/2, so flour 3 × 3/2 = 4½ cups and eggs 2 × 3/2 = 3.

Remember: Times the factor, never plus a number.

2. Inequalities ★ FOCUS

L31-9, L32-7

- "more than" → > "fewer/less than" → < "at least" → ≥ "at most / no more than / just / limited to" → ≤.
- Solve like an equation: 70m > 210 → divide both sides by 70 → m > 3.
- Test the boundary: does 3 work in m > 3? 70 × 3 = 210, which is NOT more than 210. So 3 is out, and 4, 5, 6 are in.

Example: Earn $15 per hour and want MORE than $60: 15h > 60 → h > 4. Whole numbers: 5, 6, 7.

Remember: Write the inequality, solve it, then test a number.

3. Decimals & money (multi-step) ★ FOCUS

L31-10, L31-12, L32-1, L32-11

- Underline the key words. "$23.43 EACH for 2 pairs" means 2 × 23.43 before adding.
- Adding: line up the decimal points. Copy numbers carefully (23.43, not 22.43).
- Multiplying: multiply like whole numbers, then count the decimal places in both factors (0.09 × 2.4 → 3 places → 0.216).
- Dividing by a decimal: move both decimal points until the divisor is whole (1.08 ÷ 0.9 → 10.8 ÷ 9).

Example: 3 tickets at $12.25 each + $8.40 snacks: 3 × 12.25 = 36.75; 36.75 + 8.40 = $45.15.

Remember: Estimate first: about 2 × 23 + 63 + 35 ≈ 144. Is your answer close?

4. Range, mode & mean ★ FOCUS

L31-4

- Range = largest − smallest. Mode = the value that appears most often.
- Mean = (sum of all values) ÷ (how many values). Count the values, and add twice to check.
- The mean must be between the smallest and largest values. If you get a number outside that range (or far from the middle), re-add.

Example: 80, 85, 85, 90, 95: range 15, mode 85, mean 435 ÷ 5 = 87.

Remember: Water park temperatures: sum 987, 11 days, mean ≈ 89.7.

Mini-Lessons & Tips (continued)

5. Unit rate & constant of proportionality

L31-3, L31-7, L32-8, L33-8, L33-9

- Unit rate = amount for ONE: divide (12 cups ÷ 3 pounds = 4 cups per pound).
- Constant of proportionality k = y ÷ x. It's the same in every row of a proportional table. Then y = kx.
- To fill in a table, multiply x by k: k = ⅔ and x = 4.5, so y = ⅔ × 4.5 = 3.

Example: y = 4x → k = 4, which means 4 badges per scout.

Remember: In y = kx, k is the number in front of x.

6. Fractions & complex fractions

L31-1, L32-10, L33-1, L33-2

- Subtract mixed numbers: borrow if needed. 5 − 3¾ = 4 4/4 − 3¾ = 1¼.
- Divide fractions: keep, change, flip. ⅓ ÷ ⅙ = ⅓ × 6 = 2.
- A complex fraction (fraction over fraction) is a division: (3¾)/(5/2) = 15/4 ÷ 5/2 = 15/4 × 2/5 = 3/2.

Example: (2½)/(¾) = 5/2 × 4/3 = 20/6 = 3⅓.

Remember: Change mixed numbers to improper fractions before multiplying or dividing.

7. Integers

L31-2, L31-5, L32-3, L32-12, L33-3, L33-6

- Multiplying: same signs → positive (−6 × −4 = 24); different signs → negative (−7 × 9 = −63).
- Adding on a number line: start at the first number; + positive → move right, + negative → move left.
- A number and its opposite are additive inverses: they add to 0. Gains are +, losses are −.

Example: −8 + 4: start at −8, move 4 right → −4. Same as 4 + (−8).

Remember: Count the negative signs: even → positive, odd → negative.

8. Expressions: simplify, evaluate, write

L31-8, L32-4, L32-6, L32-9, L33-4, L33-12

- Distribute first: 2(3x + 4) = 6x + 8 (the 2 multiplies BOTH terms). Then combine like terms.
- Evaluate: put the number in parentheses where the letter was: 3(−5) + 1 = −14.
- Writing: "3 less than l" → l − 3. "change from $20 after 2 drinks at s each" → 20 − 2s.

Example: 2(2a − 1) + 3 − a = 4a − 2 + 3 − a = 3a + 1.

Remember: u means 1u: 15u + u = 16u.

9. Equations

L33-7, L33-11

- Turn words into an equation: "10 is one-fifth of the total t" → ⅕t = 10 (or t ÷ 5 = 10).
- Undo in reverse order: 6n − 10 = 20 → add 10 → 6n = 30 → divide by 6 → n = 5.
- Check by putting your answer back in.

Example: ⅓p = 7 → p = 21.

Remember: Multiple choice? You can also test each choice.

Mini-Lessons & Tips (continued)

10. Volume & surface area

L31-11, L32-2, L33-5

- Volume of a box: V = l × w × h (cubic units). Missing side: V ÷ (the two known sides multiplied).
- Surface area: SA = 2(lw + lh + wh) (square units). Missing side: divide by 2, subtract the known face, then divide.
- Cube: 6 equal square faces. SA ÷ 6 = one face → edge → V = edge^3.

Example: Cube SA = 150 → face 25 → edge 5 → V = 125.

Remember: Volume → ^{3}; area → ^{2}.

Fix-It: Lesson #31 Problems to Redo

These four need another look. Read the slip, then try the problem again before reading the fix.

#4 **Water park temperatures 87, 94, 85, 91, 92, 88, 87, 91, 90, 95, 87. Find the range, mode, and mean.**

★ The slip: range (10) and mode (87) are right, but the mean is off.
Sum: 87+94+85+91+92+88+87+91+90+95+87 = 987 (add twice!).
11 days: 987 ÷ 11 = 89.7 (rounded).
Check: the mean must be in the middle of 85 and 95. 89.7 makes sense.

Answer: Range 10, mode 87, mean ≈ 89.7

#6 **Trail mix: 2 cups nuts for every 1 cup cranberries, serves 3. How much of each for 4 people?**

★ The slip: nuts were written as 3 cups (2 + 1). Adding doesn't keep the ratio.
Scale factor = 4 ÷ 3 = 4/3.
Nuts: 2 × 4/3 = 8/3 = 2⅔ cups. Cranberries: 1 × 4/3 = 1⅓ cups.
Check: 2⅔ is 2 × 1⅓ ✓ (still 2 to 1).

Answer: 2⅔ cups mixed nuts and 1⅓ cups cranberries

#9 **70 Calories per mile. How many miles to burn MORE than 210 Calories? Write an inequality, list three whole numbers.**

★



The slip: "3 miles" isn't an inequality, and 3 miles burns exactly 210, not MORE than 210.
Inequality: 70m > 210.
Divide both sides by 70: m > 3.
Whole numbers bigger than 3: 4, 5, 6.

Answer: 70m > 210 → m > 3; for example 4, 5, 6

#10 **Shoes $23.43 EACH for 2 pairs, costume $63.12, accessories $34.51. Total?**

The slips: the shoes weren't doubled, and 23.43 was copied as 22.43.
★ Shoes: 2 × 23.43 = 46.86.
Total: 46.86 + 63.12 + 34.51 = 144.49.
Estimate: 47 + 63 + 35 = 145 ✓

Answer: $144.49

Lesson #31: Test-Review Problems

Name: _____ Date: _____

Show your work in the space. A gold star under the number = focus skill.

#1 Tyson used ⅓ of a gallon of paint to cover ⅙ of the fence. At that rate, how many gallons of paint does he need to paint the entire fence?

Answer: _____

#2 Which phrase means the same as "a number and its opposite"? A) opposite of opposite B) additive inverses C) absolute value

Answer: _____

#3 The table shows a proportional relationship. The unit rate is 4 dogs per 1 cat. Write the constant of proportionality for this data.

Cats (x)	Dogs (y)
½	2
1	4
9	36
12	48

k = _____

#4 ★ Study the average temperatures recorded for each day at a water park: 87, 94, 85, 91, 92, 88, 87, 91, 90, 95, 87. Find the range, mode, and mean.

Range: _____

Mode: _____

Mean: _____

Lesson #31: Test-Review Problems (continued)

#5 Find the product. -6×-4

Answer: _____

#6 ★ A trail mix recipe calls for 2 cups of mixed nuts for every cup of cranberries and will serve 3 people. If Joachim wants to serve 4 people, how much of each ingredient will he need?

Mixed nuts: _____

Cranberries: _____

#7 A lasagna recipe calls for 12 cups of tomato sauce, 2 cups of cheese, 3 pounds of ground beef, and 1 cup of mushrooms. Give the unit rate in cups of sauce per pound of ground beef.

Answer: _____

#8 Simplify this expression: $2(2a - 1) + 3 - a$. Then evaluate it if $a = -5$.

Simplified: _____

Value when $a = -5$: _____

Lesson #31: Test-Review Problems (continued)

#9 ★ Mary Kate burns 70 Calories for every mile she runs. How many miles does she have to run to burn MORE than 210 Calories? Write an inequality where m represents a number of miles. Solve it, and list three whole numbers that make it true.

Inequality: _____

Solve: _____

Three whole numbers: _____

#10 ★ Aria entered a dance competition. She paid $23.43 EACH for 2 pairs of shoes, $63.12 for a costume, and $34.51 for accessories. What was the total?

$ _____

#11 The dimensions of a rectangular prism are 4 ft, 11 ft, and k ft. Its volume is 308 ft^3. Find the measure of the other dimension.

k = _____

#12 ★ 0.09 × 2.4 = ?

Answer: _____

Lesson #32: Test-Review Problems

Name: _____ Date: _____

Show your work in the space. A gold star under the number = focus skill.

#1 1.08 ÷ 0.9 = ?

★

Answer: _____

#2 The surface area of a cube is 486 in^2. Find the volume of the cube.

Answer: _____

#3 Find the product. −7 × 9

Answer: _____

#4 Jennifer thinks these expressions are equivalent: 2(3x + 4) and 3x + 8. Do you agree? Explain. Yes, they are equivalent No, they are not equivalent

2(3x + 4) = _____

Agree? Why? _____

Lesson #32: Test-Review Problems (continued)

#5 ★ A party punch recipe calls for one 2-liter bottle of ginger ale, one gallon of fruit juice, and one half gallon of sherbet. Lola has to cut the recipe down: she uses only ¾ gallon of fruit juice. How much of each of the other ingredients will she need?

Ginger ale: _____

Sherbet: _____

#6 Simplify the expression by combining like terms. $4x + 19 - 5 + 3x + 11x$

Answer: _____

#7 ★ Betsy's calling plan restricts her to just 50 text messages per month. Which inequality shows how many texts (x) Betsy can send and receive in a month? A) $x < 50$ B) $x > 50$ C) $x \geq 50$ D) $x \leq 50$

Answer: _____

#8 The table shows a proportional relationship. The unit rate is ⅔ burger per 1 hot dog. Complete the table and write the constant of proportionality.

Hot Dogs	Burgers
1	⅔
3	2
4.5	?
13.5	?

4.5 hot dogs → _____

13.5 hot dogs → _____

k = _____

Lesson #32: Test-Review Problems (continued)

#9 Steven bought 2 soft drinks at a cost of s dollars each. Write an expression to show how much change he got back from a $20 bill.

Answer: _____

#10 During weekly swimming lessons, Oliver increased the distance he could swim. In the first week, he swam ⅓ of the length of the pool, and by the third week, he swam 2⅔ of the length of the pool. How much farther did Oliver swim by the third week?

Answer: _____

#11 1,081 ÷ 23 = ?

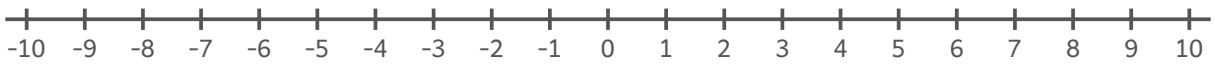
★

Answer: _____

#12 Use the number line to solve −8 + 4. Give the sum. Which of these has the same value?
A) 8 + 4 B) 8 − 4 C) 4 + (−8)

Sum: _____

Same value: _____



Lesson #33: Test-Review Problems

Name: _____ Date: _____

Show your work in the space. A gold star under the number = focus skill.

#1 Sophie helped her grandfather paint a fence. They used 5 gallons of paint. If Sophie's grandfather used 3¾ gallons, how many gallons did Sophie use?

Answer: _____

#2 Simplify this complex fraction: (3¾) ÷ (5/2)

$$\frac{3\frac{3}{4}}{5/2}$$

Answer: _____

#3 This morning, Sarah's team started at zero and earned 5 points for perfect attendance. The team lost 10 points for disruptive behavior at lunch. Write an addition equation to describe how many points Sarah's team earned. Give the sum. 0 + 5 + (−10) = −5 0 + 5 + 10 = 15 5 − (−10) = 15 0 + (−5) + 10 = 5

Equation: _____

Sum: _____

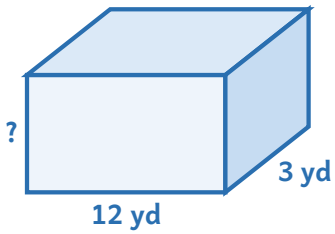
#4 Simplify this expression: 15u − 4 + u − 6. Evaluate the expression if u = 2.

Simplified: _____

Value when u = 2: _____

Lesson #33: Test-Review Problems (continued)

#5 The surface area of a rectangular prism is 342 yd^2. Two of the dimensions are 3 yd and 12 yd. What is the measurement of the other dimension?



Answer: _____

#6 When two factors have the same sign, the product is positive. Pick ALL the products that will be positive. A) $-\frac{1}{4} \times \frac{3}{4}$ B) $\frac{1}{4} \times \frac{3}{4}$ C) $\frac{1}{4} \times -\frac{3}{4}$ D) $-\frac{1}{4} \times -\frac{3}{4}$

Positive products: _____

#7 Bryan scored 10 points at last night's basketball game. This is one-fifth the total number of points scored by the entire team. Write and solve an equation to find the total number of points (t) scored by the team.

Equation: _____

Total: _____

#8 Any proportional relationship can be written as y = kx, where k is the constant of proportionality. The number of badges earned (y) is 4 times the number of scouts (x): y = 4x. How many badges will be earned per scout? What is the constant of proportionality?

Badges per scout: _____

k = _____

Lesson #33: Test-Review Problems (continued)

#9 There are 8 servings and 1,400 Calories in a container of ice cream. How many Calories are in one serving? Write the unit rate in Calories per serving.

Answer: _____

#10 ★ Iman is making cookies. The recipe calls for 6 cups of flour and 2 cups of milk. If Iman uses 4 cups of milk, how many cups of flour will she need?

Answer: _____

#11 Which of these makes the equation $6n - 10 = 20$ true? 4 5 6 7

Answer: _____

#12 The width of a desk is 3 inches less than its length. Write an expression to represent the width of the desk. Let l equal the length.

Width = _____

Practice A: Scaling Recipes

Name: _____ Date: _____

Find the scale factor (new ÷ old), then MULTIPLY every ingredient by it. Check the ratio stays the same.

1. A soup recipe uses 3 cups of broth for every 1 cup of rice and serves 4. How much of each for 6 people?

Answer: _____

2. A cookie recipe uses 2 cups of sugar and 3 eggs and makes 24 cookies. How much of each for 36 cookies?

Answer: _____

3. A salad uses 4 cups of lettuce for every 1 cup of carrots and serves 6. How much of each for 4 people?

Answer: _____

4. Lemonade uses 6 cups of water and 2 cups of lemon juice. If you use 5 cups of lemon juice, how much water?

Answer: _____

5. A recipe uses 1 gallon of juice, 3 liters of soda, and ½ gallon of sherbet. You only use ⅔ gallon of juice. How much soda and sherbet?

Answer: _____

Practice A: Scaling Recipes (continued)

6. Paint mix: 3 cans blue for every 2 cans white. How many cans of blue go with 8 cans of white?

Answer: _____

Practice B: Inequalities

Name: _____ Date: _____

more than >, fewer than <, at least ≥, at most / no more than / just ≤. Solve, then test the boundary number.

1. Jayden earns $12 for every lawn he mows. How many lawns must he mow to earn MORE than $48? Write an inequality, solve it, and list three whole numbers.

Inequality: _____

Solve / numbers: _____

2. A plant grows 3 cm per week. After how many weeks will it have grown MORE than 21 cm? Inequality + three whole numbers.

Inequality: _____

Solve / numbers: _____

3. A phone plan allows no more than 200 minutes (x). Write the inequality.

Answer: _____

4. You must be at least 48 inches tall (h) to ride. Write the inequality.

Answer: _____

5. Solve and give two whole numbers that work: $5n > 35$

Inequality: _____

Solve / numbers: _____

6. Is 3 a solution of $70m > 210$? Explain.

Answer: _____

Practice C: Decimals & Money

Name: _____ Date: _____

Underline "each" → multiply first. Line up decimal points to add. Count decimal places to multiply. Estimate to check.

1. Maya bought 3 notebooks at $4.29 EACH and a backpack for $27.50. What was the total?

Total: $ _____

2. Tickets cost $18.75 each for 4 people, plus $12.60 for parking. Total?

Total: $ _____

3. Sam paid $15.49 each for 2 shirts, $32.25 for jeans, and $8.99 for socks. Total?

Total: $ _____

4. 0.06 × 3.5 = ?

Answer: _____

5. 0.4 × 0.07 = ?

Answer: _____

6. 2.16 ÷ 0.6 = ?

Answer: _____

7. 1.45 ÷ 0.5 = ?

Answer: _____

8. 966 ÷ 21 = ?

Answer: _____

Practice D: Range, Mode & Mean

Name: _____ Date: _____

Range = biggest − smallest. Mode = most often. Mean = sum ÷ how many. Count the values and add twice!

1. Find the range, mode, and mean: 78, 85, 92, 85, 90, 88, 85

Range: _____

Mode: _____

Mean: _____

2. Find the range, mode, and mean: 12, 15, 11, 15, 18, 14, 13, 15, 20

Range: _____

Mode: _____

Mean: _____

3. Find the range, mode, and mean: 90, 94, 88, 94, 97, 91, 94, 86

Range: _____

Mode: _____

Mean: _____

4. Four quiz scores are 80, 90, 85, and 85. What score on a 5th quiz makes the mean exactly 86?

Answer: _____

Practice E: Mixed Review

Name: _____ Date: _____

Integers, fractions, unit rate, expressions, equations, volume & surface area.

1. $-9 \times -5 = ?$

Answer: _____

2. $-8 \times 6 = ?$

Answer: _____

3. $-11 + 4 = ?$

Answer: _____

4. Which equals $-3 + 9$? A) $3 - 9$ B) $9 + (-3)$ C) $-9 + 3$

Answer: _____

5. $6 - 2\frac{5}{8} = ?$

Answer: _____

6. Simplify $(2\frac{1}{2})/(\frac{3}{4})$

Answer: _____

7. ¼ gallon covers ⅛ of a deck. Gallons for the whole deck?

Answer: _____

8. $4\frac{5}{6} - 1\frac{1}{6} = ?$

Answer: _____

9. Table: x = 2, 5, 8; y = 6, 15, 24. Find k.

Answer: _____

10. $7.50 for 3 pounds. Price per pound?

Answer: _____

11. Simplify $3(2b - 4) + 5 - b$. Evaluate at $b = -2$.

Answer: _____

12. Write an expression: change from $10 after buying 3 pens at p dollars each.

Answer: _____

13. Simplify $9y + 7 - 2 + y - 4y$

Answer: _____

14. Twelve is one-third of the total t. Write and solve.

Answer: _____

15. Which makes $4n + 6 = 30$ true? 5, 6, 7, 8

Answer: _____

16. Box 5 ft × 6 ft × k ft has volume 210 ft^3. Find k.

Answer: _____

17. A cube's surface area is 294 cm^2. Find its volume.

Answer: _____

18. SA of a prism is 248 m^2. Two dimensions are 4 m and 10 m. Find the third.

Answer: _____

Answer Key

For checking after you finish. Lessons #31–#33 first, then Practice A–E.

Lesson #31

#1 ⅓ ÷ ⅙ = ⅓ × 6 = 2 gallons

#2 B) additive inverses: 5 + (−5) = 0

#3 k = 4 (every y is 4 times x, so y = 4x)

#4 Range 95 − 85 = 10. Mode 87 (three times). Mean: the sum is 987, and 987 ÷ 11 ≈ 89.7

#5 −6 × −4 = 24

#6 Factor 4/3. Nuts: 2 × 4/3 = 8/3 = 2⅔ cups. Cranberries: 1 × 4/3 = 1⅓ cups. (Nuts are still 2 × cranberries.)

#7 12 ÷ 3 = 4 cups of sauce per pound of beef

#8 4a − 2 + 3 − a = 3a + 1. 3(−5) + 1 = −15 + 1 = −14

#9 70m > 210, so m > 3. Whole numbers that work: 4, 5, 6 (any whole number bigger than 3; 3 miles burns exactly 210, not more).

#10 2 × 23.43 = 46.86; 46.86 + 63.12 + 34.51 = $144.49

#11 44k = 308 → k = 7 ft

#12 9 × 24 = 216 → 0.216

Lesson #32

#1 10.8 ÷ 9 = 1.2

#2 486 ÷ 6 = 81 = 9 × 9, so the edge is 9 in. V = 9 × 9 × 9 = 729 in^3

#3 −7 × 9 = −63

#4 2(3x + 4) = 6x + 8, which is not 3x + 8. So no, Jennifer is wrong (try x = 1: 14 vs 11).

#5 Factor ¾. Ginger ale 2 × ¾ = 1½ liters. Sherbet ½ × ¾ = ⅜ gallon.

#6 18x + 14

#7 D) x ≤ 50 (at most 50, and 50 itself is allowed)

#8 ⅔ × 4.5 = 3; ⅔ × 13.5 = 9; k = ⅔

#9 20 − 2s

#10 2⅔ − ⅓ = 2⅓ pool lengths

#11 1,081 ÷ 23 = 47 (23 × 47 = 1,081)

#12 −8 + 4 = −4. C) 4 + (−8) = −4 too (you can add in any order).

Lesson #33

#1 5 − 3¾ = 4 4/4 − 3¾ = 1¼ gallons

#2 15/4 × 2/5 = 30/20 = 3/2 = 1½

#3 0 + 5 + (−10) = −5 points

#4 16u − 10; 16(2) − 10 = 22

#5 2(36 + 15x) = 342 → 36 + 15x = 171 → 15x = 135 → x = 9 yd

#6 B (both positive) and D (both negative). Each equals 3/16.

#7 ⅕t = 10 (or t ÷ 5 = 10), so t = 50 points

#8 4 badges per scout; k = 4

#9 1,400 ÷ 8 = 175 Calories per serving

#10 Factor 4 ÷ 2 = 2, so 6 × 2 = 12 cups of flour

#11 6n = 30, so n = 5 (6 × 5 − 10 = 20 ✓)

Answer Key (continued)

#12 I – 3

Practice A: Scaling

1. 4 1/2, 1 1/2 (factor 6/4 = 3/2: broth 3 × 3/2 = 4½ cups, rice 1½ cups)
2. 3, 4 1/2 (factor 36/24 = 3/2: sugar 3 cups, eggs 4½)
3. 2 2/3, 2/3 (factor 4/6 = 2/3: lettuce 2⅔ cups, carrots ⅔ cup)
4. 15 (factor 5/2: 6 × 5/2 = 15 cups (or 3 water per 1 juice))
5. 2, 1/3 (factor ⅔: soda 3 × ⅔ = 2 L, sherbet ½ × ⅔ = ⅓ gal)
6. 12 (factor 8/2 = 4: 3 × 4 = 12 cans)

Practice B: Inequalities

1. $12L > 48 \rightarrow L > 4$; e.g. 5, 6, 7
2. $3w > 21 \rightarrow w > 7$; e.g. 8, 9, 10
3. $x \leq 200$
4. $h \geq 48$
5. $n > 7$; e.g. 8, 9
6. No: 70 × 3 = 210, and 210 is not greater than 210

Practice C: Decimals & money

1. $40.37 (3 × 4.29 = 12.87; 12.87 + 27.50 = 40.37)
2. $87.60 (4 × 18.75 = 75.00; + 12.60 = 87.60)
3. $72.22 (2 × 15.49 = 30.98; 30.98 + 32.25 + 8.99 = 72.22)
4. 0.21 (6 × 35 = 210 → 3 places → 0.210)
5. 0.028 (4 × 7 = 28 → 3 places → 0.028)
6. 3.6 (21.6 ÷ 6 = 3.6)
7. 2.9 (14.5 ÷ 5 = 2.9)
8. 46 (21 × 46 = 966)

Practice D: Range, mode, mean

1. range 14, mode 85, mean 603/7 ≈ 86.1
2. range 9, mode 15, mean 133/9 ≈ 14.8
3. range 11, mode 94, mean 367/4 ≈ 91.8
4. 90 (5 × 86 = 430; 430 − 340 = 90)

Practice E: Mixed review

1. 45
2. −48
3. −7
4. B (= 6)
5. 3⅜
6. 10/3 = 3⅓
7. 2
8. 3⅔

Answer Key (continued)

9. $k = 3$ ($y = 3x$)
10. $2.50
11. $5b - 7$; -17
12. $10 - 3p$
13. $6y + 5$
14. $t/3 = 12 \rightarrow t = 36$
15. 6
16. 7 ft
17. 343 cm^3
18. 6 m

For grown-ups

• Focus skills come from the Lesson #31 problems that need another look: #4 (mean), #6 (recipe scaling), #9 (inequality word problem), #10 (multi-step money). #12's answer in the photo was too faint to check.
• The interactive page works offline, reads problems aloud, gives instant same-skill practice after a miss, and retests missed problems until they're right.